

A Quantity Theory of Money Approach to Cryptocurrency Valuation: Store of Value, Medium of Exchange, and the Economic Moat Signal

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Abstract

We apply the Quantity Theory of Money (QTM) to the cross-sectional valuation of cryptocurrencies and protocol tokens. Treating each digital asset as the currency of a self-contained economy, we decompose market capitalization into a Medium of Exchange (MoE) component, priced by the free-circulating velocity of supply, and a Store of Value (SoV) residual. The key innovation is λ —the fraction of total supply that is locked or staked—which partitions the circulating supply into economically active and inert components. We derive a “free velocity” (V_{free}) applicable only to unlocked supply, yielding a cleaner estimate of the MoE component and, by subtraction, the SoV premium. For native blockchain coins, high λ reflects a convenience yield and

seigniorage signal grounded in monetary search theory. For governance and utility tokens, high λ carries a distinct meaning: it is an economic moat signal, capturing token holder conviction in platform growth and foreclosing the exit option. We construct two summary metrics—the NVT/Growth ratio and the SoV/MoE ratio—and arrange assets in a quadrant framework that identifies undervalued, overvalued, and moat-endowed assets. Using a panel of $[N]$ major cryptocurrencies and tokens over [sample period], we find that quadrant positioning significantly predicts future returns, with “Star” assets (high SoV/MoE, low NVT/G) generating superior risk-adjusted returns over 12–24 month horizons.

1 Introduction

What determines the value of a cryptocurrency? The dominant approach in the literature has been to adapt discounted cash flow (DCF) models, searching for a dividend-like flow—transaction fees, staking rewards, governance rights—that can be capitalized into a present value (Liu and Tsyvinski, 2021). This approach, however, mischaracterizes the fundamental nature of digital assets. Cryptocurrencies and protocol tokens are not equity claims on future earnings. They are currencies. Each coin or token is the monetary instrument of a self-contained digital economy, and its value is determined by the same forces that determine the value of any currency: the scale of economic activity it enables and the efficiency with which it circulates. The correct framework, we argue, is the Quantity Theory of Money (Fisher, 1911).

The Fisher equation, $MV = PQ$, relates the stock of money (M) and its velocity (V) to the nominal value of economic output (PQ) (Fisher, 1911). Applied to digital assets, a cryptocurrency’s market capitalization proxies its money supply, and the economic output it supports—on-chain transaction volume for native coins, protocol throughput for tokens—is its PQ . Under this mapping, a coin’s market capitalization satisfies $MC \approx PQ/V$, making value a direct function of economic activity and circulation efficiency. This is not merely an analogy. Cong, Li, and Wang (2021) develop a formal dynamic model in which token prices aggregate heterogeneous users’ transactional demand, with equilibrium prices governed by platform adoption—a formulation directly consistent with the QTM framework we develop here. Sockin and Xiong (2020) similarly model tokens as platform membership instruments, showing that the token price reflects the present value of the convenience yield accruing to participants: the non-pecuniary benefit of network membership, including access to platform services, network liquidity, and future transactional capability. We explain the role of this convenience yield in our valuation decomposition below.

Before proceeding, we define the two components of value our framework decomposes. The *Medium of Exchange* (MoE) value of a digital asset is the value attributable to its

current transactional function: the economic activity it enables today, priced by the velocity with which it circulates. The *Store of Value* (SoV) value is the premium agents are willing to pay today by forgoing current exchange utility—the transactions, goods, services, or alternative investments they could obtain by spending the asset now—in anticipation of future value appreciation. This appreciation is expected to arise from network growth, expanding user adoption, increasing utility, and innovation. An agent holding Ethereum rather than spending it is implicitly paying an intertemporal price: she sacrifices today’s exchange utility for her conviction that the network’s future worth exceeds today’s use value. The SoV component is the aggregate market valuation of this forward-looking conviction.

The QTM framework requires a critical extension to operationalize this decomposition. Not all of a cryptocurrency’s circulating supply is economically active. A substantial and growing fraction is locked in staking contracts, pledged as governance collateral, or held in long-term custody with near-zero velocity. These holdings are economically inert from a transactional standpoint: they contribute to the money supply but perform no current exchange function. Conflating active and inactive supply distorts velocity estimates and, consequently, misprices the economic content of market capitalization. We address this by introducing λ , defined as the proportion of the total token supply that is locked or staked:

$$\lambda = \frac{M_{\text{locked}}}{M}.$$

The free-circulating supply is then $(1 - \lambda)M$, and we define the *free velocity* V_{free} as the velocity of only this active portion. This yields the MoE component of market capitalization as

$$\text{MC}_{\text{MoE}} = \frac{PQ}{V_{\text{free}}},$$

and, by residual, the Store of Value component as

$$\text{MC}_{\text{SoV}} = \text{MC} - \text{MC}_{\text{MoE}}.$$

MC_{SoV} captures everything the market assigns to the asset beyond its current transactional utility: the aggregate premium holders pay today by voluntarily forgoing current exchange utility. The ratio $MC_{\text{SoV}}/MC_{\text{MoE}}$ is our primary measure of the balance between current transactional utility and forward-looking appreciation premium.

Why does λ matter beyond its mechanical role in adjusting velocity? For native blockchain coins—Ethereum, Solana, and their peers—high λ is grounded in three complementary theoretical traditions. First, *monetary search theory* (Kiyotaki and Wright, 1993) establishes that money emerges as a coordination equilibrium: an asset becomes valuable because rational agents expect others to hold and accept it. High λ is the blockchain-observable, publicly verifiable signal of this coordination commitment. Unlike traditional monetary systems where holding preferences are latent, on-chain data makes coordination commitments transparent and measurable. Second, the *convenience yield* (Sockin and Xiong, 2020; Krishnamurthy and Vissing-Jorgensen, 2012) is the non-pecuniary benefit agents derive from holding a monetary asset: access to network security, platform liquidity, and future transactional capability. Just as holders of U.S. Treasury securities accept lower yields in exchange for the safety and liquidity services of dollar-denominated assets (Krishnamurthy and Vissing-Jorgensen, 2012; Nagel, 2016), holders of dominant blockchain coins accept lower required returns because network membership itself has value beyond current transactions. Crucially, the SoV premium is the present value of expected *future* convenience yields: agents who lock coins today are paying for anticipated growth in the network’s value and their membership in it. High λ therefore signals that a large share of holders expect the network’s convenience yield to expand, making λ a forward-looking indicator of perceived network quality. Third, the *seigniorage* channel (Cong and He, 2024) anchors this signal in rational optimization: in proof-of-stake networks, newly minted tokens are distributed to stakers as block rewards—a direct analog to monetary seigniorage, where the issuer transfers value from non-stakers to stakers through new issuance. By choosing to stake, an agent reveals that expected seigniorage income plus anticipated price appreciation exceeds her opportunity cost of lock-

ing capital—a revealed preference for the network’s positive net present value. These three channels are complementary: coordination theory explains why high λ is self-reinforcing, convenience yield provides the holding motive, and seigniorage grounds the rational optimization underlying the staking decision.

Governance and utility tokens present a related but distinct case. A governance token such as UNI is not the settlement currency of Ethereum’s transaction economy; it is the currency of the Uniswap protocol economy. Its PQ is not on-chain gas consumption but the total throughput of the Uniswap protocol—decentralized exchange volume, liquidity deployed, fees generated. Whether protocol revenue currently accrues to token holders is immaterial to this characterization, just as it is immaterial whether U.S. GDP “flows to” dollar holders. The dollar has value because the U.S. economy is large and growing. UNI has value because the Uniswap economy is large and growing. This framing aligns with Cong, Li, and Wang (2021), who price tokens on the present value of future platform utility and adoption. For governance tokens, however, λ carries a different signal than for coins. Staking a governance token into a DAO contract renders it illiquid: the staked tokens cannot be sold while locked. An agent who voluntarily incurs this illiquidity cost reveals a preference for future appreciation over present liquidity—a choice she would not make unless she holds conviction that the protocol’s economy will grow. This conviction is grounded in assessments of network effects, utility expansion, competitive positioning, and innovation trajectory; it is fundamental rather than speculative. Drawing on Hirschman (1970), holders who exercise *voice* (governance participation) rather than *exit* (selling) reveal a preference for the protocol’s future that passive buyers do not. Edmans and Manso (2011) formalize this in a corporate setting: blockholders who engage rather than exit create governance value precisely because engagement is costly and therefore credible. Applied to tokens, active governance participation—staking plus voting—is a costly signal (Spence, 1973) of fundamental conviction in platform growth, and high participation rates are an economic moat indicator. Recent empirical evidence confirms this: DAO Governance (2024) document that proposal

passage increases DAO token returns by 4.7%, and that a one-standard-deviation increase in voting participation amplifies this effect by 2.2%.

We operationalize these ideas in two summary metrics and an investment framework. The *NVT/Growth ratio* (NVT/G) normalizes the Network Value to Transactions ratio by the network’s growth rate, yielding a PEG-ratio analog for digital assets: assets with low NVT/G are cheap relative to their adoption trajectory. The *SoV/MoE ratio* measures the balance between forward-looking appreciation premium and current transactional utility, with high values indicating strong fundamental conviction in future network growth. Plotting assets on a 2×2 matrix across these dimensions yields four quadrants with distinct investment implications. *Star/Economic Moat* assets (low NVT/G, high SoV/MoE) are cheap relative to growth and carry strong holder conviction—the best quadrant and the primary long signal in our empirical tests. *GARP* assets (low NVT/G, low SoV/MoE) offer utility-driven value at reasonable prices with modest forward-looking premium. *Overvalued* assets (high NVT/G, high SoV/MoE) are expensively priced relative to growth yet retain high holder conviction; they are not necessarily to be avoided but are priced for perfection, and represent a potential transition opportunity if conviction materializes into adoption. *Avoid* assets (high NVT/G, low SoV/MoE) combine expensive pricing relative to growth with weak holder conviction—the worst combination and the primary short signal. We find that Star assets generate [X]% higher annual returns than Avoid assets over [horizon], after controlling for size, momentum, and market beta.

Related Literature. Our paper contributes to several strands of the literature. The first is the theory and empirics of cryptocurrency valuation. Cong, Li, and Wang (2021) develop a dynamic model of token adoption and valuation grounded in network externalities; Sockin and Xiong (2020) model tokens as platform membership with convenience yield pricing; Liu and Tsyvinski (2021) document systematic risk factors in cryptocurrency returns. We complement these by providing an empirically tractable decomposition of market capitalization

grounded in QTM, and by constructing testable investment signals from on-chain observables. The second strand is the monetary economics of digital assets. Kiyotaki and Wright (1993) and subsequent search-theoretic models of money inform our treatment of λ as a coordination signal; Cong and He (2024) analyze the tokenomics of proof-of-stake staking; the convenience yield literature (Krishnamurthy and Vissing-Jorgensen, 2012; Nagel, 2016) motivates the monetary premium embedded in MC_{SoV} ; Jiang, Krishnamurthy, and Lustig (2024) and the reserve currency literature provide the analogy between the dollar’s exorbitant privilege and the dominant-network premium in crypto. The third strand is DAO governance and token holder behavior. Edmans and Manso (2011) formalize voice versus exit in a blockholder setting; DAO Governance (2024) and DAO Review (2025) document empirical patterns in DAO voting and their asset pricing consequences. Our paper is the first, to our knowledge, to connect governance participation rates to a QTM-based valuation decomposition and to use them as an investment signal.

The remainder of the paper is organized as follows. Section ?? develops the formal theoretical framework, derives the SoV/MoE decomposition, and presents formal propositions distinguishing coins from governance tokens. Section ?? states the testable hypotheses. Section ?? describes the data, sample construction, and variable measurement. Section ?? presents the main empirical results. Section ?? reports robustness tests. Section ?? concludes.

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